

Impact Crater Equations

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1 Introduction

This is a compilation of several versions of the Maxwell Z model equations and other relevant equations for impact crater formation. Repeated equations across papers are not listed. SI units unless otherwise specified. Some equations refer specifically to conditions on the Moon, specifically the diameter at the simple to complex transition = 15km on the Moon.

2 Melosh 2011

2.1 Maxwell Z Equations (Box 6.1)

$$u_r = \frac{\alpha(t)}{rZ} \quad (1)$$

$$u_\theta = (Z - 2)u_r \frac{\sin \theta}{1 + \cos \theta} \quad (2)$$

$$r = s(1 - \cos \theta)^{\frac{1}{Z-2}} \quad (3)$$

$$H_{exc} = \frac{D_{at}}{2}(Z - 2)(Z - 1)^{\frac{1-Z}{Z-2}} \quad (4)$$

$$Z = 3 \implies H_{exc} = \frac{D_{at}}{8} \approx \frac{H_{at}}{3} \quad (5)$$

$$M_{ej} = \frac{Z - 2}{Z - 1}M_e \quad (6)$$

$$u_V = \frac{\alpha}{sZ} \quad (7)$$

$$u_H = (Z - 2)u_V \quad (8)$$

2.2 Ejecta Distribution (ch. 6.4)

”[In simple craters, t]he thickness of the ejecta falls off as roughly the inverse cube of distance from the rim”

$$\text{Diameter} < 15\text{km} \implies \delta(r) \approx (r - R)^{1/3} \quad (9)$$

$$\delta(r) = f(R) \left(\frac{r}{R} \right)^{-3 \pm 0.5} \quad (10)$$

$$f(R) \text{ is constant} \implies \delta(r) = y \cdot R \left(\frac{r}{R} \right)^{-3 \pm 0.5} \quad (11)$$

2.3 Scaling of Crater Dimensions (ch. 6.2.2, ch. 6.3.2 and ch. 6.5)

$$D > D_{s-c} \text{ and } D_{s-c} = 15\text{km} \implies D = 0.823 D_{at}^{1.13} \quad (12)$$

$$H_{exc} = 1/3 \cdot H_{at} \quad (13)$$

$$H_{exc} = 1/10 \cdot D_{at} \quad (14)$$

$$\frac{D}{D_0} = \left(\frac{W}{W_0} \right)^{1/3} \quad \text{*exponent is } \frac{1}{3.4} \text{ in larger explosions} \quad (15)$$

$$\frac{H}{H_0} = \left(\frac{W}{W_0} \right)^{1/3} \quad \text{*exponent is } \frac{1}{3.4} \text{ in larger explosions} \quad (16)$$

$$D_{at} = 1.161 \left(\frac{\rho_p}{\rho_t} \right)^{1/3} L^{0.78} v_i^{0.44} g^{-0.22} \sin^{1/3} \theta_i \quad (17)$$

2.4 Constants

- Z : the parameter of the Maxwell Z model. It falls between 2 and 4. $Z = 2.7071$ conserves the total vertical momentum in the total flow field (neglecting gravity).
- s : a constant that is different for each streamline
- D_0 : final diameter of a known crater
- W_0 : explosive energy that yielded a known crater
- H_0 : final depth of a known crater
- g : surface gravity
- D_{s-c} : the diameter at which craters transition from simple to complex (15km on the Moon)

2.5 Variables and Functions

- u_r : the radial component of the of the excavation flow velocity
- $\alpha(t)$: a function of time that describes the strength of flow
- u_θ : the angular component of the flow velocity in polar coordinates
- (r, θ) : polar coordinate parameters
- $(r(t), \theta(t))$: the time-dependent polar position of a point that originated at (r_0, θ_0)
- H_{exc} : maximum depth of excavation
- M_{ej} : total mass of ejecta
- M_e : total mass displaced from the transient crater
- u_V : vertical velocity component
- u_H : horizontal velocity component
- D_{at} : transient crater diameter
- H_{at} : transient crater depth
- R_{at} : transient crater radius
- $\delta(r)$: ejecta thickness as a function of radial distance from the crater center
- D : final crater diameter
- W : explosive energy that yielded a crater
- H : final crater depth
- ρ_t : target density
- ρ_p : projectile density
- L : projectile diameter
- v_i : impact velocity
- θ_i : impact angle

3 Maxwell 1977

3.1 Equations

$$\nabla \cdot \mathbf{u} = 0 \quad (18)$$

$$r^{Z+1}(t) = r_0^{Z+1} + (Z+1) \int_0^t \alpha(t) dt \quad (19)$$

$$\alpha \text{ is constant} \implies r^{Z+1} = r_0 + (Z+1)\alpha t \quad (20)$$

$$r_c^{Z+1}(t) = (Z+1) \int_0^t \alpha(t) dt \quad (21)$$

$$\alpha \text{ is constant} \implies r_c^{Z+1} = (Z+1)\alpha t \quad (22)$$

Note: constant α and Z are not compatible with conservation of energy, but can be a good approximation.

$$R = \left[\frac{4Z(Z-2)\alpha^2}{g} \right]^{\frac{1}{2Z+1}} \quad (23)$$

$$R_2 = R[1 + \frac{1}{2}Z] \quad (24)$$

$$R = K' \left[\frac{4Z(Z-2)E^{2Z/3}}{g} \right]^{\frac{1}{2Z+1}} \quad (25)$$

$$Z = 2.7071 \implies R = \frac{K \cdot E^{1/3.55}}{g^{1/6.4}} \quad (26)$$

3.2 Constants

- α : the function $\alpha(t)$ (see Melosh) can be taken as constant as an approximation
- K' : constant which depends on explosive type, explosive placement, and the cratering material type
- K : $\approx 5 \pm 0.5$ when energy is expressed as equivalent metric tons of TNT, gravity in m/s/s, and R in meters

3.3 Variables and Functions

- R : final crater radius
- R_2 : hinge point of the flap
- E : kinetic energy release

4 Other Equations

4.1 Diameter and Depth Ratios

$$D > 15\text{km} \implies H_{exc} \in [\sim 0.05D, \sim 0.1D] \quad (27)$$

$$\frac{H}{D} \approx 0.1 \quad (28)$$

4.2 Ejecta Thickness

Austin Equation:

$$\delta(r) = 0.14 \pm 0.062 \cdot R^{0.77 \pm 0.080} \left(\frac{r}{R} \right)^{-2.8 \pm 0.1} \quad (29)$$